

where $e_{\gamma,\rho}(x_1, t_1, x_2, t_2) = -\infty$ for any other choice of x_1, x_2 . Then the metric e arising from planting measures along a finite or countable set of curves is the smallest metric which is bounded below by satisfying $e \geq d$, and for each curve γ_i and planted measure ρ_i , $e \geq e_{\gamma_i, \rho_i}$. All metrics not of this form have infinite rate.

For each such planted measure, the contribution to the rate function is $I_i = \frac{4}{3} \int \rho_i^{\frac{3}{2}} dt$, so that the rate function I of a metric e is the sum over all planted measures $I = \sum_i I_i$. Our goal in this paper is to find the lowest rate metric for which a geodesic F from $(0, 0)$ to $(0, 1)$ passes through the point $(1, a)$.

Our metrics will generally have a measure planted along just one curve, which will be the geodesic from $(0, 0)$ to $(0, 1)$. Any measure planted outside of this geodesic will not increase its length, but may increase the length of paths other than F , which only reduces the possibility of F being a geodesic, all while increasing the rate.

We can then define e or $e(F, \rho)$ as the metric arising from planting ρ along F , which will be our geodesic.

We will in fact prove a result about e : Suppose $\mathcal{M}_\varepsilon$ is a metric sampled from the conditional law of $\mathcal{L}_\varepsilon$, given that the rightmost geodesic γ of e from $(0, 0)$ to $(0, 1)$ satisfies $\gamma(a) \geq 1$.

Theorem 1.3. *As $\varepsilon \rightarrow 0$, $\mathcal{M}_\varepsilon \rightarrow e(F, \rho)$ in probability with respect to uniform convergence on bounded sets, with F as in Theorem 1.2 and ρ as given by:*

$$\rho(t) = \begin{cases} \frac{1}{a^2(3-\sqrt{8a})^2}, & \text{if } t \leq 2a \\ \frac{2}{at(3-\sqrt{8a})^2}, & \text{if } 2a \leq t \leq 1. \end{cases}$$

2 Solution of a relaxed problem

One possible candidate for F is two straight line segments, from $(0, 0)$ to $(1, a)$ and from $(1, a)$ to $(0, 1)$, with ρ just sufficient to compensate d , such that $e(0, 0, F(t), t) = 0, \forall t \in [0, 1]$. However, for $a < \frac{1}{2}$, this will not be a geodesic: instead, there will be straight lines from $(0, 0)$ to a point beyond $(1, a)$ whose Dirichlet length is greater than the length of F to that point. As a result, the geodesic will skip the steeper line segment to $(1, a)$ and take such a straight line as a shortcut to F , and then follow part of the other line segment to $(0, 1)$. Similarly, if $a > \frac{1}{2}$, it will follow part of the line segment from $(0, 0)$ to $(1, a)$ and then shortcut the steep segment to proceed directly to $(0, 1)$.

This suggests that we should try to solve for a metric where the segment from $(0, 0)$ to $(1, a)$ is not skipped by such a shortcut. In other words, we should ask what is the lowest rate metric for which there is a geodesic candidate F , so that the distance from $(0, 0)$ along F to any point on F is greater than the distance of a shortcut from $(0, 0)$ to that point.

Inspired by this intuition, we will first solve an intermediate optimization problem: fix F_0 and ρ_0 (and therefore $e(F_0, \rho_0)$) for $t \in [0, t_1]$. Consider pairs (F, ρ) that agree with (F_0, ρ_0) on $[0, t_1]$. Which pair minimizes the rate of the